C — Deliverable 3 Document

Project Title: Library Management System
Course: CS3009 — Software Engineering
Deliverable: Assignment 3 — Final Report
Team Name: Stack Architects

Team Members:

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1. Introduction

The Library Management System (LMS) is a full-stack web application designed to automate and streamline library operations for students, librarians, and administrators.

- **Sprint 1:** Established account management, including student registration, admin approval, role-based login, and account deletion.
- **Sprint 2:** Added book catalog, search, borrowing workflow, approval/rejection, due date assignment, borrow history, and borrow limit enforcement.
- **Sprint 3:** Completed the remaining advanced modules: book returns, automatic fine calculation, fine management, renewal/reissue, reservation/waitlist system, and admin reports. Sprint 3 also included **final integration, UI improvements, and finishing touches** to ensure a complete and fully functional system.

By the end of Sprint 3, the LMS became a **fully integrated library management solution**, covering the entire lifecycle from registration to borrowing, returning, fines, renewal, reservations, and reporting.

2. Project Vision

The vision of the LMS is to provide a **secure, efficient, and user-friendly platform** for library management. Sprint 3 enhanced the system to:

- Automate return and overdue fine processing.
- Enable students to renew and reserve books.
- Provide librarians with comprehensive fine and reservation management.

- Give administrators accurate reports for library activity.
- Ensure the system is fully integrated, consistent, and ready for deployment.

3. Intended Use of the System

Students:

- Search books, request loans, track active loans.
- Renew loans and reserve unavailable books.
- View fines and payment status.

Librarians:

- Manage books, approve/reject borrow requests, record returns.
- Track overdue books, manage fines, and handle reservation queues.

Administrators:

- Approve or reject accounts, manage user roles.
- View reports on issued books, overdue loans, fines, and system activity.

Stakeholder Updates:

- No new stakeholders were added in Sprint 3.
- Student, Librarian, and Admin workflows were enhanced to support the new modules.

4. Features and Functionality

Sprint	**Features Completed**
Sprint 1	Registration, admin approval/rejection, role-based login, account deletion request
Sprint 2	Book catalog management, search, borrow request workflow, approval/rejection, due date assignment, borrow history, borrow limits
Sprint 3	Book Return Module, Automatic Fine Calculation, Fine Management, Renewal/Reissue, Reservation/Waitlist, Admin Reports, Final Integration and UI polishing

Sprint 3 Enhancements:

- Completed the borrowing lifecycle with returns, fines, and renewal modules.
- Ensured fair allocation with reservation queue management.
- Provided administrators with real-time reporting.
- Integrated all modules for a smooth end-to-end workflow.

5. User Stories (Sprint 3)

- As a Librarian, I want to record book returns so that issued books become available.
- As a System, I want to calculate overdue fines automatically so that penalties are applied accurately.
- As a Student, I want to view my fines so that I can check my outstanding balance.
- As a Librarian, I want to mark fines as paid or waived to maintain accurate records.
- As a Student, I want to renew borrowed books so that I can extend my loan period.
- As a Student, I want to reserve unavailable books to receive them when returned.
- As a Librarian, I want to manage reservation queues so that books are assigned fairly.
- As an Admin, I want to view overdue, issued, fine, and activity reports to monitor library operations.

6. Non-Functional Requirements (NFRs)

Performance:

- The system shall respond to page requests within 2–3 seconds.
- Return, renewal, and reservation operations shall execute without delays.

Usability:

- User dashboards shall be intuitive and clearly display available actions.
- Success and error messages shall appear after relevant actions.

Reliability:

- Duplicate returns, fines, or reservations shall be prevented.
- Inventory and user account data shall remain consistent.

Security:

- Only authenticated users may access protected pages.
- Access to Admin, Librarian, and Student pages shall be restricted by role.
- CSRF protection is enabled for all POST actions.

Scalability:

- Database design supports multiple users, books, borrow records, fines, and reservations.
- The system is designed to accommodate future module expansions and reporting features.

7. Testing and Quality Assurance

Test Plan:

- Focused on Return, Fine, Renewal, Reservation, and Admin Reports modules.
- Functional, regression, UI, and integration testing were conducted.
- Test environment: Spring Boot 3.5, Java 17, MySQL, Thymeleaf, localhost:8081.

Black-Box Test Cases (10):

- Loan return, duplicate return, overdue fine calculation, student fine view, fine management, renewal allowed and blocked scenarios, reservation creation, admin report accuracy.

White-Box Test Cases (10):

- Branches and conditions for return, fine calculation, fine management, renewal rules, reservation creation and READY/EXPIRED logic.

Bug Report Summary:

- All critical Sprint 3 defects were identified and fixed.
- Key fixes included duplicate return validation, fine calculation corrections, renewal rule enforcement, and reservation queue management.

Test Results and Observations:

- 100% of selected black-box and white-box tests passed.
- Regression tests confirmed no disruption of Sprint 1 or 2 functionality.
- System correctly handles all end-to-end workflows for Sprint 3 modules.

How Testing Improved Quality:

- Ensured module logic correctness, data consistency, and adherence to business rules.
- Validated system integration across all three sprints.

8. Iteration 3 Implementation Screenshots

9. Work Division for Sprint 3

Ali Bin Salman:

- Led final integration and Sprint 3 planning
- Developed return and fine management requirements
- Verified dashboards and route access
- Coordinated documentation and report preparation

Muhammad Rohaan:

- Implemented backend logic for return, fine, renewal, and reservation modules
- Ensured transactional safety and service-layer logic
- Developed admin reports queries
- Maintained Scrum board and tracked task progress

Muhammad Atyab:

- Designed and implemented Sprint 3 UI pages
- Performed black-box and white-box testing
- Captured screenshots for the report
- Verified bug fixes and regression testing

Collaborative Work:

- Regression testing across all three sprints
- Integration of Sprint 3 modules with previous features
- Final UI polishing and defect fixing

- Documentation preparation for deliverable submission

10. Final System Summary

Across all three sprints, the LMS evolved into a **complete, integrated system**:

- **Sprint 1:** Account management and authentication
- **Sprint 2:** Book catalog, search, borrowing workflow, and history
- **Sprint 3:** Return, fine, renewal, reservation, admin reporting, and final integration

The system now provides end-to-end support for library operations: registration, borrowing, return, fines, renewal, reservations, and administrative monitoring. It is stable, fully functional, and aligned with the original project scope.